BUSINESS SYSTEM OPTIMIZATION TECHNIQUES

UNIT I

FOUNDATIONS OF BUSINESS SYSTEMS & ENTERPRISE PROCESSES

DR. SUCHITRA AGRAWAL

2 OVERVIEW

- Before we learn optimization, we must understand what exactly we are trying to optimize.

3 WHAT IS A SYSTEM?

- A system is a set of **interrelated components** working together to achieve a **common goal**.
- Each component has their own role.
- Failure of one affects the whole system.
- Input–Process–Output

4 SYSTEM ELEMENTS

- Input: resources, data, people
- Process: transformation activities
- Output: products, services, results

- Examples: College, Railway, Human Body, Computer System, Hospital
- Feedback: Poor results -> helps in improving things

- Transformation logic
- Control and feedback

5 BUSINESS SYSTEM

- When a system is designed to achieve business objectives, it is called a business system.
- People, processes, technology
- Goal-oriented
- Dynamic and open
- Examples: Factory, IT Company, Hospital
- A business is not random—it works as a system. Optimization improves this system.

6 CHARACTERISTICS

- Goal-oriented: Every system has objectives
- Interdependent: Departments depend on each other
- Dynamic: Changes with time and environment
- Feedback-driven: Uses results to improve

7 OPEN VS CLOSED SYSTEMS

- Open: environmental interaction, continuous interaction. Example: Business organization
- Closed: minimal interaction, no interaction with environment. Example: Pressure Cooker
- Environmental Factors could be: Customers, Government, Market Competition

8 IMPORTANCE OF WELL DEFINED BUSINESS SYSTEM

- Efficiency: ability of a business system to achieve desired output with minimum use of resources.
- Coordination: integration of activities of different departments to achieve organizational goals.
- Decision support: refers to the use of information systems to assist managers in making informed decisions.
- Businesses survive only because they adapt to environmental changes.
- Business systems are open and dynamic, so optimization is a continuous process.

9 ENGINEERING ORGANIZATIONS

- Engineering organizations are companies whose operations are based on engineering knowledge, engineering projects or engineering products.
- Manufacturing firms → mechanical/electrical engineering
- IT companies → software engineering
- Infrastructure firms → civil engineering
- Energy & utilities → power engineering
- Service sectors

- Organizations are not one big blob. They are made up of multiple smaller systems called subsystems.

10 EXAMPLE

- When a car is produced → mechanical + electrical + industrial + project engineering
- When a metro line is constructed → civil + electrical + signaling + IT + project management

11 ROLE OF ENGINEERS

- Engineers are problem solvers. In organizations, their role is not just to build, but to improve the way things work.
- Process design
- System integration
- Optimization
- Automating operations
- Improving productivity
- Ensuring quality and reliability

- Optimization is literally an engineering approach applied to business.

12 INTERNAL ENVIRONMENT

- The internal environment includes everything **within the organization's control**.
- Structure
- Culture
- People
- Resources
- Policies

13 EXTERNAL ENVIRONMENT

- External environment includes all factors **outside the organization's control** that affect it.
- Customers
- Competitors
- Government Regulations
- Economy
- Suppliers
- Technology Trends
- Social trends

14 STAKEHOLDERS

- Stakeholders are individuals or groups that:
  → **affect** or are **affected by** organizational decisions.
- Employees
- Managers
- Owners
- Customers
- Suppliers
- Government
- Society
- Creditors
- Shareholders

15 SYSTEM BOUNDARIES

- Defines scope
- Helps modelling

- When doing blood test, you check blood — not whole body.
- When analyzing **production system**:
  - Include: Machines, Workers, Materials
  - Exclude: Customer Buying Behaviour

16 SUBSYSTEMS

- Operation / Production
- Finance
- HR
- Marketing and Sales
- Research and Development
- Supply Chain / Logistics

- Each subsystem has its own goals and processes, but works toward the organizational goal.

17 INTERDEPENDENCY

- Subsystems cannot work in isolation. If one subsystem fails, the whole organization feels the impact.
- Subsystem influence
- Need for coordination

18 EXAMPLES

- Example 1: Hiring delay
  - HR can't hire skilled workers → production slows → orders delayed → customers unhappy
- Example 2: Finance delay
  - Finance delays payment → vendors don't supply → production stops
- Example 3: Sales & production mismatch
  - Sales sells more than production can deliver → customer dissatisfaction
- Example 4: R&D & marketing disconnect
  - R&D builds a product no one wants → marketing failure

19 TECHNOLOGY ENABLEMENT

- Technology improves how business systems operate. It supports:
  - Automation
  - Digital systems
  - Integration
  - Data Flow
  - Speed
  - Accuracy

20 ORGANIZATIONAL STRUCTURES

- There are two common ways organizations are structured:
- Functional
- Process-oriented

21 FUNCTIONAL STRUCTURE

- A functional structure organizes people based on specialized business functions like Finance, HR, Marketing, and Operations.
- Examples:
  - Banks
  - Traditional Firms
  - Govt. Ministry System
- Features:
  - Department-based
  - Specialization

SAMPLE STEPS UNDER FUNCTIONAL STRUCTURE

- Sales team gets a requirement
- Production team manufactures
- Finance bills the client
- HR manages manpower
- Marketing promotes product

23 ADVANTAGES OF FUNCTIONAL STRUCTURE

- Specialization: People develop deep expertise.
- Economies of Scale: Centralized resources reduce cost.
- Clear Authority: Reporting relationships are defined.
- Simplified Training: Same skills in one group → easier training.

24 FUNCTIONAL STRUCTURE LIMITATIONS

- Departmental Silos: Departments focus on their goals, not company goals.
- Poor Coordination: Departments hand work to each other slowly.
- Slow decision making: Information must move up and down hierarchy.
- Customer Unfocused: Structure focuses on **tasks**, not **value delivery**.
- Local Optimization vs Global Optimization: Departments optimize themselves, not whole organization.

25 PROCESS-ORIENTED STRUCTURE

- Organizes work around core processes that deliver value to the customer, instead of grouping by functions.
- The flow here is horizontal — it follows the journey of delivering value.
- Examples:
  - Amazon
  - Toyota
  - Infosys
  - McDonald's
- Features:
  - Horizontal Flow
  - Cross-Functional teams
  - Shared Accountability
  - Customer centric

26 SAMPLE STEPS UNDER PROCESS-ORIENTED STRUCTURE

- Customer places order
- Order is confirmed
- Product/service is produced
- Product is shipped
- Customer is billed
- Payment is received
- Customer is supported

- This entire chain involves multiple functions but is seen as **one integrated process**.

ADVANTAGES OF PROCESS-ORIENTED STRUCTURE

- Faster Coordination
- Faster response
- Eliminates redundancy
- Improves transparency
- Supports continuous improvement
- Easier to optimize

28 COMPARISON

Feature	Functional	Process-Oriented
Orientation	Internal tasks	Customer value
Structure	Vertical	Horizontal
Focus	Department goals	End-to-end process
Coordination	Weak	Strong
Optimization	Local	System-wide
Speed	Slow	Faster
Flexibility	Low	High
Ownership	Department manager	Process owner
Customer view	Indirect	Direct

IMPACT OF STRUCTURE ON PERFORMANCE

- Functional structures were suitable for stable environments. Whereas, process structures are suitable for dynamic, competitive environments.
- Efficiency
- Speed
- Quality

30 BUSINESS PROCESS

- A business process is a sequence of activities that takes inputs and transforms them into outputs that deliver value to a customer or stakeholder.
- Example: Supermarket Process:
  - Customer gives product
  - Cashier scans
  - System updates inventory
  - Bill generated
  - Payment received
  - Inventory updated

Inputs → activities → output → value

31 BUSINESS PROCESS EXAMPLES

✓ **Manufacturing:**

Order → Production Planning → Assembly → Packaging → Dispatch

✓ **Banking:**

Account Opening → KYC → Approval → Activation → Customer Support

✓ **IT Services:**

Requirement Gathering → Development → Testing → Deployment → Maintenance

✓ **Food Chain (McDonald's):**

Order Taking → Cooking → Assembly → Delivery → Billing

✓ **E-commerce (Amazon):**

Search → Order → Payment → Fulfilment → Shipping → Tracking → Delivery

PROCESS TYPES

- Core: Directly deliver value to customers. Eg: production, sales, delivery, service
- Support: Enable or assist core processes. Eg: HR, Accounting, IT Support, Procurement
- Management: Govern & control the organization. Eg: Planning, Budgeting, Strategic Decisions, Compliance

33 CHARACTERISTICS

- Goal Oriented
- Cross Functional
- Customer Centric
- Measurable
- Repeatable
- Optimizable

Processes determine the actual performance of the organization.

34 VALUE CHAIN

- A sequence of activities that adds value at each step, from raw material to delivered product.
- For example: Steps for a T-Shirt selling process would be:
  - Cotton harvested
  - Yarn spun
  - Cloth woven
  - Cloth dyed
  - T-shirt stitched
  - Branding + packaging
  - Retail Sale

PORTER'S VALUE CHAIN MODEL

- Support Activities
- Primary Activities
- Margin

36 PRIMARY ACTIVITIES

Directly involved in creating and delivering the product.

- Inbound Logistics
  - Warehousing
  - Raw Material Storage
  - Inventory Control
- Operations
  - Machining
  - Assembly
  - Manufacturing
  - Packaging
- Outbound Logistics
  - Distribution
  - Transportation
  - Order Fulfilment
- Marketing and Sales
  - Pricing
  - Advertising
  - Promotion
  - Sales Channel
- Service
  - Maintenance
  - Repair
  - Customer Service

37 SUPPORT ACTIVITIES

Enable primary activities to work better.

- Firm Infrastructure
  - Finance
  - Accounting
  - Legal
  - Strategic Planning
- HR
  - Hiring, Training
- Procurement
  - Purchasing of materials/equipment
- Technology
  - R&D
  - Process Automation
  - IT Systems
  - Innovation

OPTIMIZATION VIEW

- Cost reduction for non-value adding services
- Increases efficiency in value addition services

39 INFORMATION SYSTEMS

- An information system collects, processes, stores and distributes information to support operations, management and decision-making.

Information System = Data + Process + Technology + People

TPS: TRANSACTION PROCESSING SYSTEM

- Records and processes routine transactions of the organization.
- Characteristics:
  - Repetitive
  - Structured
  - High Volume
  - Accuracy focused
  - Real time or batch processing

41 TPS EXAMPLES

- Payroll processing
- Attendance recording
- Billing systems
- Order Entry
- Inventory updates
- ATM transactions
- Ticket booking

- TPS supports **operational level** employees and processes.

MIS: MANAGEMENT INFORMATION SYSTEM

- Converts TPS data into summarized information for managers.
- Characteristics:
  - Summarization
  - Reporting
  - Exception Alerts
  - Periodic Updates

43 MIS EXAMPLES

- Monthly Sales Reports
- Inventory Usage Reports
- Expense Summaries
- Exception Reports (Overbudget)

- Supports **middle management** for planning, monitoring & controlling.

44 TPS VS MIS

Parameter	TPS	MIS
Level	Operational	Managerial
Data	Detailed	Summarized
Frequency	Continuous	Periodic
Function	Record transactions	Assist decisions
Output	Receipts/Updates	Reports

DSS: DECISION SUPPORT SYSTEM

- Supports semi-structured and unstructured decisions using analytical models, data and simulations.
- Tools used by DSS:
  - Models
  - Simulations
  - Optimization
  - Forecasting
  - Scenario Analysis

DSS EXAMPLES

- Forecasting Demand
- Inventory Planning
- Pricing Decisions
- Scheduling and Routing
- Capacity Planning

- Supports **tactical decision making**.

47 ESS: EXECUTIVE SUPPORT SYSTEM

Provides top executives with summary, trend and forecasting information to support strategic decisions.

Used for:

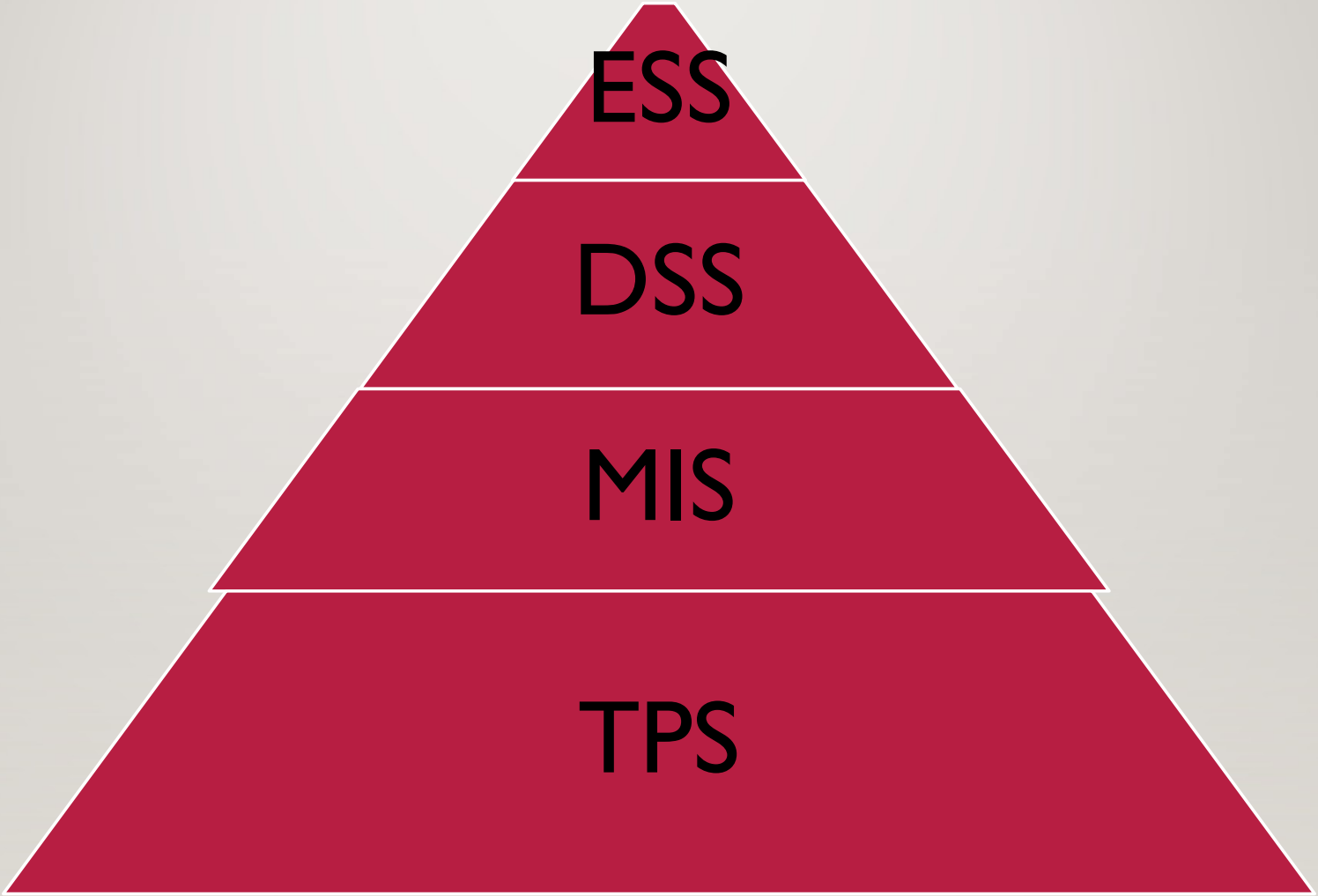
- Long Term Planning
- Market Entry Decisions
- Investment Decisions
- Competitive Analysis

Uses:

- Dashboards
- Scorecards
- Trend Graphs
- KPIs

ESS EXAMPLE

- CEO dashboard showing information about:
  - Revenue trends
  - Sales regions
  - Profit Margins
  - Stock Performance
  - Competitor Benchmarks
- Supports **strategic decision makers** (top management)



As you move up the pyramid, data becomes more summarized and decisions become more unstructured.

50 ENTERPRISE SYSTEMS

- TPS, MIS, DSS, and ESS support decision-making within departments. But organizations need systems that connect departments together. These are called Enterprise Systems.
- Enterprise Systems = ERP + CRM + SCM

51 ERP: ENTERPRISE RESOURCE PLANNING

- ERP integrates core business functions into a single unified system.
- Before:
  - Each department used its own system
  - Data was duplicated
  - Errors were common
  - Decision-making was slow

ERP Solves and Integrates all the functionalities into one central database.

ERP MODULES

- ✓ Financial Management
- ✓ Human Resources
- ✓ Procurement
- ✓ Inventory Management
- ✓ Production Planning
- ✓ Sales & Distribution
- ✓ Order Management

ERP EXAMPLES

- ✓ SAP ERP
- ✓ Oracle ERP
- ✓ Microsoft Dynamics
- ✓ Tally Prime (Indian SME market)
- ✓ Odoo (modular open-source ERP)

ERP BENEFITS

- ✓ Eliminates data duplication
- ✓ Real-time visibility
- ✓ Improves coordination
- ✓ Speeds decision-making
- ✓ Reduces delays
- ✓ Supports optimization

CRM: CUSTOMER RELATIONSHIP MANAGEMENT

- CRM manages interactions with customers to improve acquisition, retention and satisfaction.
- Functions:

✓ track leads and opportunities

✓ record customer interactions

✓ manage sales pipeline

✓ manage complaints & service requests

✓ analyse customer buying patterns

In Porter's value chain → CRM covers:

- Marketing & Sales
- Service

56 CRM EXAMPLES

- Salesforce CRM
- Zoho CRM
- HubSpot
- FreshSales
- Microsoft Dynamics CRM

Use Cases:

- Tracking Customer Complaints
- Managing Sales Leads
- Offering Personalized Discounts
- Forecasting Sales
- Segmenting Customers

CRM supports customer retention via personalization, loyalty programs, customer support

SCM: SUPPLY CHAIN MANAGEMENT

- SCM manages the flow of materials, information and finances from supplier to customer.
- Stages:
  - Supplier Management
  - Procurement
  - Manufacturing
  - Warehousing
  - Transportation
  - Distribution
  - Retail/Delivery

SCM BENEFITS

- ✓ reduced inventory cost
- ✓ reduced lead time
- ✓ improved delivery performance
- ✓ reduced stock-outs
- ✓ demand forecasting

LINKING ERP, CRM AND SCM

Suppliers → SCM → ERP → CRM → Customers

✓ SCM → handles supply side

✓ CRM → handles customer side

✓ ERP → integrates internal functions

SDLC: SYSTEM DEVELOPMENT LIFE CYCLE

- SDLC is a structured process for developing, implementing and maintaining information systems.
- Ensures:
  - Proper Planning
  - Correct Development
  - Minimal Errors
  - User Satisfaction
  - Quality Assurance

61 SDLC PHASES (WATERFALL MODEL)

- Phase 1: Requirement Analysis
- Phase 2: System Design
- Phase 3: Development (Coding)
- Phase 4: Testing
- Phase 5: Deployment / Implementation
- Phase 6: Maintenance

- Sequential Model: Output of one phase becomes input to the next.

62 REQUIREMENTS ENGINEERING

- Requirement Engineering is the process of identifying, analyzing, specifying and validating user requirements.
- Activities:
  - Requirement Elicitation
  - Requirement Analysis
  - Requirement Documentation
  - Requirement Validation

REQUIREMENT TYPES

- Functional Requirements
  - What the system should do?
- Non-Functional Requirements
  - How the system should behave?

DOCUMENTATION

- Documentation records how processes, systems, and procedures work in the organization.
- If a process is not documented, it is difficult to understand, measure, or improve.
- Clarity
- Completeness

OPTIMIZATION ROLE

- Optimization improves performance of business processes to achieve better efficiency, cost, time, and quality.

✓ reduce delays

✓ remove waste

✓ lower cost

✓ improve customer satisfaction

✓ increase throughput